

Autonomous Vehicle Trajectory Optimization under Friction Constraints

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1 Introduction and Problem Statement

This paper outlines the design and validation of an emergency lateral lane-change trajectory for an autonomous vehicle traveling at high speed. The primary objective is to ensure that the generated path strictly respects tire-road friction thresholds and passenger comfort criteria to prevent vehicle slip and maintain directional stability.

1.1 Operational Parameters

The maneuver is defined by the following kinematic and geometric constraints:

- **Longitudinal Velocity (v_x):** 25 m/s (≈ 90 km/h, assumed constant during the maneuver).
- **Lateral Displacement (D):** 3.5 m (National highway lane width in india).
- **Target Maximum Lateral Acceleration ($a_{y,\max}$):** ≤ 4.0 m/s² (Safety limit on typical dry asphalt to prevent skidding).
- **Target Maximum Lateral Jerk ($j_{y,\max}$):** ≤ 2.0 m/s³ (Passenger comfort threshold to prevent violent steering snap).

2 Mathematical Foundation

If a path $y(x)$ possesses a discontinuous curvature, the steering wheel would be required to instantaneously teleport from the center to a turned position. This discontinuous jump results in infinite lateral jerk ($\dot{a}_y = \infty$), which would destabilize the vehicle's suspension and cause an immediate loss of control.

To ensure smooth steering engagement, we enforce boundary conditions not just for position, but also for heading (first derivative) and curvature (second derivative). We utilize a quintic (5th-degree) polynomial to provide the necessary degrees of freedom:

$$y(x) = c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + c_5x^5 \quad (1)$$

2.1 Boundary Conditions

To solve for the six unknown coefficients $[c_0, c_1, c_2, c_3, c_4, c_5]^T$, we define six boundary conditions at the start ($x = 0$) and the end ($x = X_f$) of the maneuver:

2.1.1 Initial State ($x = 0$)

- Initial Position: $y(0) = 0$
- Initial Heading (Straight): $y'(0) = 0$
- Initial Curvature (Zero Steering Angle): $y''(0) = 0$

2.1.2 Final State ($x = X_f$)

- Final Position (Target Lane): $y(X_f) = D$
- Final Heading (Aligned with Lane): $y'(X_f) = 0$
- Final Curvature (Straightened Steering): $y''(X_f) = 0$

3 Translating Physics to Linear Algebra

To map these physical constraints into a computationally solvable format, we take the derivatives of our quintic polynomial with respect to longitudinal distance x :

$$y'(x) = c_1 + 2c_2x + 3c_3x^2 + 4c_4x^3 + 5c_5x^4 \quad (2)$$

$$y''(x) = 2c_2 + 6c_3x + 12c_4x^2 + 20c_5x^3 \quad (3)$$

Applying the initial boundary conditions at $x = 0$ yields immediate values for the first three coefficients:

$$\begin{aligned} y(0) &= c_0 = 0 \\ y'(0) &= c_1 = 0 \\ y''(0) &= 2c_2 = 0 \implies c_2 = 0 \end{aligned}$$

Substituting these known zero-coefficients and evaluating the remaining system at the final longitudinal distance $x = X_f$, we construct the exact linear system $A \cdot C = B$:

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & X_f & X_f^2 & X_f^3 & X_f^4 & X_f^5 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 2X_f & 3X_f^2 & 4X_f^3 & 5X_f^4 \\ 0 & 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 2 & 6X_f & 12X_f^2 & 20X_f^3 \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ c_2 \\ c_3 \\ c_4 \\ c_5 \end{bmatrix} = \begin{bmatrix} 0 \\ D \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad (4)$$

By reducing the system to the three remaining non-trivial equations, the matrix equation simplifies cleanly to:

$$\begin{bmatrix} X_f^3 & X_f^4 & X_f^5 \\ 3X_f^2 & 4X_f^3 & 5X_f^4 \\ 6X_f & 12X_f^2 & 20X_f^3 \end{bmatrix} \begin{bmatrix} c_3 \\ c_4 \\ c_5 \end{bmatrix} = \begin{bmatrix} D \\ 0 \\ 0 \end{bmatrix} \quad (5)$$

Solving this system analytically or numerically via a linear solver yields the unique trajectory polynomial coefficients optimized for the specified lane-change maneuver length X_f .

4 Computational Implementation

The following complete, object-oriented Python implementation structures the trajectory generation, maps the parametric search space, and verifies kinematic parameters against both safety and comfort metrics.

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 class TrajectoryGenerator:
5     def __init__(self, lane_width=3.5, velocity=25.0):
6         self.D = lane_width
7         self.vx = velocity
8         self.C = None # Stores polynomial coefficients
9
10    def solve_coefficients(self, xf):
11        x0 = 0
12        A = np.array([
13            [1, x0, x0**2, x0**3, x0**4, x0**5], # y(0)
14            [1, xf, xf**2, xf**3, xf**4, xf**5], # y(xf)
15            [0, 1, 2*x0, 3*x0**2, 4*x0**3, 5*x0**4], # y'(0)
16            [0, 1, 2*xf, 3*xf**2, 4*xf**3, 5*xf**4], # y'(xf)
17            [0, 0, 2, 6*x0, 12*x0**2, 20*x0**3], # y''(0)
18            [0, 0, 2, 6*xf, 12*xf**2, 20*xf**3] # y''(xf)
19        ])
20        B = np.array([0, self.D, 0, 0, 0, 0])
21
22        self.C = np.linalg.solve(A, B)
23        return self.C
24
25    def verify_safety(self, xf):
26        x_vals = np.linspace(0, xf, 100)
27        c3, c4, c5 = self.C[3], self.C[4], self.C[5]
28
29        # Spatial-to-temporal transformation via chain rule
30        ay_vals = (self.vx**2) * (6*c3*x_vals + 12*c4*x_vals**2 + 20*c5*x_vals
31        **3)
32        jerk_vals = (self.vx**3) * (6*c3 + 24*c4*x_vals + 60*c5*x_vals**2)
33
34        max_ay = np.max(np.abs(ay_vals))
35        max_jy = np.max(np.abs(jerk_vals))
36
37        is_safe = max_ay <= 4.0
38        is_comfortable = max_jy <= 2.0
39
40        return is_safe, max_ay, is_comfortable, max_jy
41
42 # planning module
43 generator = TrajectoryGenerator(lane_width=3.5, velocity=25.0)
44
45 # Numerical parametric analysis loop (40m to 80m)
46 xf_range = np.linspace(40, 80, 200)
47 max_accelerations = []
48
49 for xf in xf_range:
50     generator.solve_coefficients(xf)
51     is_safe, max_ay, is_comfortable, max_jy = generator.verify_safety(xf)
52     max_accelerations.append(max_ay)
53
54 # Validation Plot
55 plt.figure(figsize=(10, 6))
56 plt.plot(xf_range, max_accelerations, label='Trajectory Max $a_y$', color='blue',
57          linewidth=2)
```

```

56 plt.axhline(y=4.0, color='red', linestyle='--', linewidth=2, label='Tire
    Friction Limit ($4.0 \text{ m/s}^2$)')
57
58 # Map the computed boundary intersection (56.99m)
59 critical_xf = 56.99
60 plt.axvline(x=critical_xf, color='green', linestyle=':', linewidth=1.5, label='
    Min Safe Distance Threshold (~57m)')
61 plt.fill_between(xf_range, max_accelerations, 4.0, where=(xf_range <
    critical_xf), color='red', alpha=0.15, label='Danger Zone (Skidding)')
62 plt.fill_between(xf_range, max_accelerations, 4.0, where=(xf_range >=
    critical_xf), color='green', alpha=0.15, label='Safe Zone (Smooth)')
63
64 plt.title('Parametric Sensitivity Analysis: Lane Change Distance vs. Peak
    Lateral Acceleration', fontsize=11, fontweight='bold')
65 plt.xlabel('Maneuver Longitudinal Distance ($X_f$) [meters]', fontsize=10)
66 plt.ylabel('Peak Lateral Acceleration ($a_{y,max}$) [$\text{m/s}^2$]', fontsize=10)
67 plt.grid(True, linestyle=':', alpha=0.6)
68 plt.legend(loc='upper right', fontsize=10)
69 plt.savefig('parametric_analysis.png', dpi=300)
70 plt.show()

```

5 Conclusion and Autonomous Control Implications

By implementing a 5th-degree quintic polynomial trajectory planner, this study successfully established a continuous mathematical profile for position, heading, and curvature across a high-speed highway lane-change maneuver. Enforcing boundary conditions up to the second derivative ensures that lateral acceleration begins and ends at exactly zero. This foundational constraint guarantees that the trajectory avoids instantaneous steering-wheel steps that would otherwise yield infinite lateral jerk ($\dot{a}_y = \infty$), preventing severe suspension destabilization and maintaining deterministic vehicle control.

The parametric sensitivity analysis conclusively illustrates the highly non-linear dynamics governing vehicle maneuvers at speed. Because peak lateral acceleration scales inversely with the square of the longitudinal distance ($a_{y,max} \propto X_f^{-2}$), minor alterations to the planned maneuver window yield disproportionately large reductions in the lateral forces required from the tire patches.

Under a strict physical threshold of $a_{y,max} \leq 4.0 \text{ m/s}^2$ —representing the realistic boundary of static friction before initiating a tire skid on typical dry asphalt—the parametric sweep mapped a definitive operational threshold. For an autonomous vehicle traveling at a constant longitudinal velocity of 25 m/s ($\approx 90 \text{ km/h}$) executing a standard 3.5 m (*indianNH*) lateral displacement, the system transitions from an unstable tracking state to a physically viable regime at a critical intersection point. The numerical model reveals that the absolute minimum safe longitudinal distance required to complete this emergency maneuver without breaking tire adhesion is **56.99 meters**.

In a production-level autonomous driving architecture, this trajectory validation script functions as a real-time predictive safety governor within the local motion-planning stack. If perception layers (such as LiDAR, Radar, or vision-based neural networks) detect an obstacle, the behavior layer can instantaneously evaluate the remaining longitudinal clearance against this computed boundary. If the spatial clearance falls below the critical 56.99 m threshold, the planning engine can immediately flag a swerving trajectory as physically impossible (predicting a catastrophic vehicle skid) and dynamically command an alternative fallback safety behavior, such as maximum straight-line emergency braking.